Root Cause Analysis Report

Part	MS-4505-001 Sheet Metal Bracket
Material / process	SPCC Steel / Bending
Case	7a3fcaac-ef46-46d6-b54f-2672da567b5b
Date	2026-11-01 18:04:34
Status	Reviewed by expert (expert)

WARNING: the AI model was not used for this analysis (fallback-fake (gemini failed: HTTP 429: rate limited / quota exceeded)). The text below is a generic offline template, not a result.

1. Problem

Defect	Severity	Confidence	Detail
scratch	minor	92%	
dent	major	88%	
crack	critical	76%	
deformation	major	91%	

- L2: -0.11 mm
- H1: -0.05 mm

2. Root cause

Cause: Improper Bending Force / Over Bending

Reason: The detected deformation and dimensional inaccuracy strongly suggest excessive or improperly calibrated bending force. Retrieved RCA documents confirm this pattern for SPCC steel bending operations.

Effect: Deformation, Dimensional Inaccuracy

Confidence: 68%

Supporting evidence
- **587aa665-c6f1-48df-8783-09d65ff4f82a**: Tool Wear / Die Damage Root Cause Analysis Defect Types: scratch, dent, edge burr, dimensional shift Process: bending, stamping, forming Root Cause: Progressive wear of punch/die tooling causes dimensional drift, surface scratches, edge burrs and dents on formed parts. Effect: Surface defects (scratches from worn tooling edges), dents from damaged die surfaces, edge burrs from worn cutting edges,
- **586f086d-0deb-40c2-a71d-f840aaa55a60**: Improper Bending Force / Over Bending Root Cause Analysis Defect Types: deformation, dimensional inaccuracy Process: sheet metal bending, press brake operations Root Cause: Improper bending force settings or over-bending beyond the material's elastic limit. Effect: Deformation of the part, dimensional inaccuracy in bent features, springback issues causing part to not meet dimensional tolerances. C
- **6872e1ff-9d0f-41f8-9cd9-c783d66c1b46**: Springback Root Cause Analysis Defect Types: deformation, dimensional inaccuracy Process: sheet metal bending Root Cause: Elastic recovery of sheet metal after forming (springback) causes the bent angle to open up beyond the intended angle. Effect: Dimensional inaccuracy in bent angles, parts out of tolerance on angular dimensions and related linear dimensions. Corrective Actions: 1. Apply overben

3. Solution

Corrective actions (fix this occurrence)

1. Recalibrate press brake
2. Verify material lot

Preventive actions (stop it recurring)

1. Schedule monthly calibration

2. Incoming inspection

Precautions

- Optimize bending parameters to material specification
- Use proper tooling and punch/die sets
- Regular machine calibration and force verification

How to verify the fix

Re-inspect the next 30 parts; no deformation and all dimensions within tolerance

4. Other causes considered

Rank	Cause	Confidence	Reason
2	Material Inconsistency / Hardness Variation	65%	Hardness variation in the incoming material can cause unpredictable springback and cracking. Material certificate verification is required.
3	Tool Wear / Die Damage	62%	Progressive die wear causes scratches, dents, and edge burrs. Preventive maintenance schedule should be implemented.

5. Traceability

AI model: fallback-fake (gemini failed: HTTP 429: rate limited / quota exceeded). Expert edited the cause: no.
This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.